

Mandatory Assignment 2

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Exercise 1

1a: Variable selection and sample restriction

First, we remove observations before year 2000. We then select ten stock-level characteristics based on the variable-importance rankings reported in [Gu et al. \(2020\)](#), Figure 5. Figure 5 shows that characteristics related to price trends, liquidity, size, and volatility consistently rank among the most important predictors across a wide range of machine learning models. Therefore, we include the price-trend variables `mom1m` and `mom12m`, the size and liquidity measures `mvel1` and `turn`, and the risk and volatility measures `beta` and `retvol`. In addition, we include the value characteristics `bm` and `ep`, the profitability measure `roaq`, and the investment measure `invest`, which receive relatively high importance rankings in Figure 5 and represent distinct return-predictive channels documented in the asset pricing literature. Rather than selecting only the highest-ranked characteristics, we aim to capture multiple dimensions of return predictability, including momentum, size, liquidity, risk, value, profitability, and investment. Together, these variables provide exposure to several of the most prominent return-predictive characteristics identified in both the machine learning and asset pricing literature.

For the macroeconomic state variables from [Goyal and Welch \(2008\)](#) library, we follow [Gu et al. \(2020\)](#) Table 4. The table shows that aggregate book-to-market `bm`, the Treasury bill rate `tbl`, the default yield spread `dfy`, and the term spread `tms` are among the most influential macroeconomic variables across several machine learning specifications. We therefore select `bm`, `tbl`, `dfy`, and `tms` as our four macroeconomic predictors. These variables capture aggregate valuation conditions, short-term interest rates, credit market conditions, and the slope of the yield curve, respectively, and are widely used in empirical asset pricing as indicators of the overall economic environment.

1b: Summary Statistics and Missing Values

Table [1](#) reports summary statistics for the selected predictors.

Table 1: Summary statistics for selected predictors

	mean	std	25%	50%	75%
characteristic_mom1m	-0.004	0.604	-0.550	-0.009	0.544
characteristic_mom12m	-0.007	0.590	-0.526	-0.026	0.519
characteristic_mvell	0.026	0.588	-0.496	0.044	0.544
characteristic_turn	0.036	0.576	-0.453	0.074	0.532
characteristic_beta	0.076	0.548	-0.370	0.102	0.540
characteristic_retvoll	0.123	0.528	-0.306	0.146	0.576
characteristic_bm	-0.065	0.531	-0.512	-0.067	0.373
characteristic_ep	-0.011	0.558	-0.486	-0.004	0.454
characteristic_roaq	-0.014	0.560	-0.477	-0.023	0.457
characteristic_invest	-0.006	0.533	-0.427	-0.000	0.413
macro_bm	0.276	0.072	0.239	0.284	0.328
macro_tbl	0.018	0.019	0.001	0.011	0.028
macro_dfy	0.010	0.004	0.008	0.009	0.012
macro_tms	0.022	0.014	0.009	0.023	0.035

We first inspect missing values across the selected stock-level characteristics, macroeconomic variables, the prediction target, lagged market capitalization, and industry classification. In the current sample, no missing values are observed in these variables, and the cleaning procedure therefore does not reduce the number of observations.

As a safeguard and to ensure a consistent preprocessing pipeline, missing values in stock-level characteristics are handled using monthly cross-sectional median imputation. This approach would preserve stock-month observations while relying only on contemporaneous information from the same month. The median is preferred to the mean because it is less sensitive to extreme firm-level observations, which are common in financial data.

Macroeconomic variables are common to all firms within a given month and vary only over time. Since no missing macroeconomic values are observed in the current sample, no macro imputation is required. Nevertheless, the preprocessing pipeline applies forward filling if missing macroeconomic values are encountered, so that only previously observed macroeconomic information is used. Backward filling is avoided to prevent look-ahead bias.

Finally, observations with missing excess returns or lagged market capitalization would be dropped if present because `ret_excess` is the prediction target and `mktcap_lag` is required to construct value-weighted portfolio returns.

After cleaning, the final sample contains 1,111,605 stock-month observations across 10,692 firms and 264 months. The missing-value treatment balances data retention with robustness, while the industry recoding keeps the dimensionality of the feature matrix manageable.

1c: Feature matrix construction

First, we recode `sic2` into the 10 major SIC divisions to retain broad industry information while keeping the feature space manageable. This reduces the number of industry dummy variables and lowers the risk that sparse industry categories create noisy predictors in the machine learning models.

Then, we construct the feature matrix using the selected stock characteristics and their interactions with macroeconomic state variables. The feature matrix contains 59 predictors: 10 stock characteristics, 40 characteristic-macro interactions, and 9 industry dummies. The interaction terms allow firm-level return predictability to vary with macroeconomic conditions, consistent with [Gu et al. \(2020\)](#). One SIC division is omitted as the reference category when constructing industry dummies.

Exercise 2

Exercise 2a: Model descriptions

Elastic Net is a penalized linear regression model that combines the variable selection property of the Lasso with the coefficient shrinkage of Ridge regression. It is well suited to our setting because the feature matrix contains many correlated predictors. The model minimizes the penalized mean squared error

$$\hat{\beta} = \arg \min_{\beta} \frac{1}{n} \|y - X\beta\|_2^2 + \lambda \left[\alpha \|\beta\|_1 + \frac{1 - \alpha}{2} \|\beta\|_2^2 \right]$$

The first term measures prediction error, while the second term combines the Lasso (L_1) and Ridge (L_2) penalties. In the sklearn implementation, the tuning hyperparameters are alpha, which controls the overall regularization strength, and l1_ratio, which determines the balance between Lasso and Ridge.

Random Forest is a nonlinear tree-based ensemble model. It builds many regression trees on different bootstrap samples of the data and averages their predictions. Each tree recursively splits the feature space into groups with similar realized excess returns. Averaging across trees reduces the instability of individual decision trees and helps improve out-of-sample performance.

The key hyperparameters are the number of trees (`n_estimators`), the number of predictors considered at each split (`max_features`), the minimum number of observations required in a leaf (`min_samples_leaf`), and tree depth (`max_depth`). Compared with Elastic Net, Random Forest is more flexible because it can capture nonlinearities and interactions without imposing a linear functional form.

Together, the two models provide a comparison between a regularized linear benchmark and a flexible nonlinear machine learning model.

Exercise 2b: Time-respecting data split

The dataset is split chronologically to avoid look-ahead bias. Training observations precede validation observations, which in turn precede test observations. The most recent 20% of months are reserved for the test set and are only used for final out-of-sample evaluation, as shown in [Table 2](#).

The remaining 80% of months is split into training and validation samples, yielding an approximate 60/20/20 split. The training set is used to estimate model parameters, while the validation set is

used for hyperparameter tuning. The split is performed by month rather than by individual rows to preserve the panel structure.

Table 2: Time-respecting train/validation/test split with date ranges and sizes

	Sample	Start date	End date	Months	Observations
0	Train	2000-01-01	2013-02-01	158	748902
1	Validation	2013-03-01	2017-07-01	53	194460
2	Test	2017-08-01	2021-12-01	53	168243

Continuous predictors are standardized using the training sample only. The same transformation is then applied to the validation and test sets, which avoids data leakage from future observations.

Exercise 2c: Hyperparameter tuning

Hyperparameters are tuned using grid search. Each candidate specification is fitted on the training sample and evaluated on the validation sample using RMSE, consistent with the prediction-error objective in [Gu et al. \(2020\)](#).

For Elastic Net, the grid varies the overall penalty strength (`alpha`) and the balance between Lasso and Ridge regularization (`l1_ratio`). For Random Forest, we use a computationally constrained grid and focus tuning on tree depth, while keeping the number of trees, feature subsampling rule, and minimum leaf size fixed. This allows us to examine how increasing model complexity affects predictive accuracy while keeping computation feasible.

Figure Figure 1 shows how training and validation RMSE vary across the tuned hyperparameters. For Elastic Net, the lines are almost overlapping, indicating limited sensitivity to `l1_ratio` in this grid. For Random Forest, deeper trees slightly improve training fit, while validation RMSE remains nearly flat.

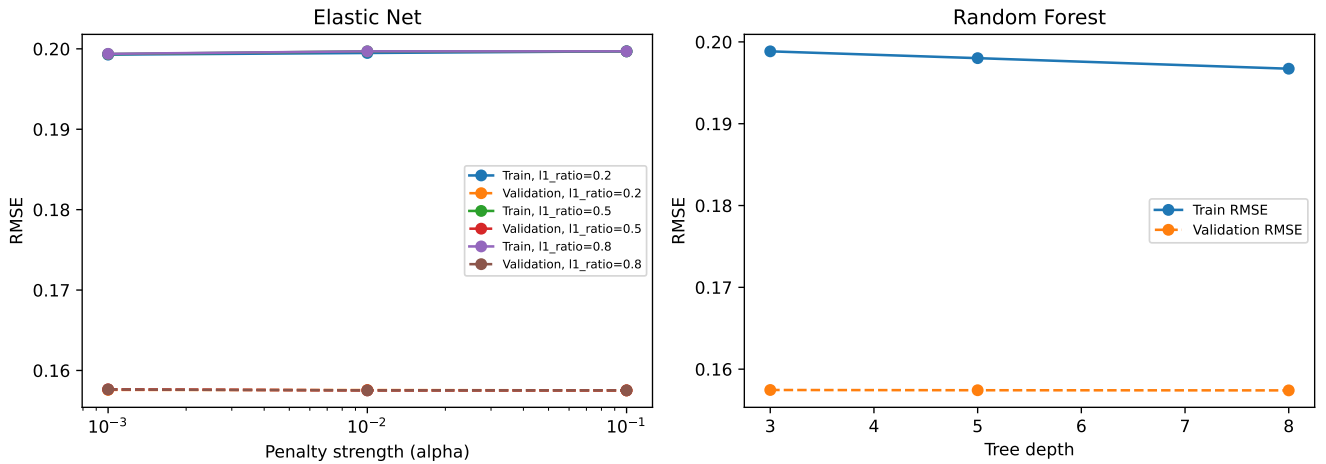


Figure 1: Training and validation RMSE across model complexity

The final specification for each model is selected as the one with the lowest validation RMSE, as reported in Table 3. Comparing training and validation RMSE illustrates the bias-variance trade-off:

more flexible models may fit the training data better, but this does not necessarily translate into better validation performance.

Table 3: Selected hyperparameters and validation RMSE

Model	Selected specification	Train RMSE	Validation RMSE
Elastic Net	alpha=0.1, l1_ratio=0.2	0.199689	0.157523
Random Forest	trees=50, depth=8, leaf=100	0.196722	0.157406

Exercise 2d: Save models to disk

To load the models for peer review, run:

```
final_en = joblib.load("model_elastic_net.joblib")
final_rf = joblib.load("model_random_forest.joblib")
```

The final fitted models are saved to disk using `joblib` to ensure reproducibility and facilitate peer review.

Exercise 3

Exercise 3a-c: Predictions and long-short portfolios

Using the fitted Elastic Net and Random Forest models from Exercise 2, we generate predicted excess returns for each stock-month observation in the validation and test samples. The test set is only used for final out-of-sample evaluation and is not used for model fitting or hyperparameter selection.

For each month, stocks are sorted into decile portfolios based on predicted excess returns. Decile 0 contains the lowest predicted returns and decile 9 contains the highest predicted returns. Realized excess returns within each decile are computed as value-weighted averages using `mktcap_lag`.

The long-short strategy buys the top predicted-return decile and shorts the bottom predicted-return decile. The monthly long-short return is therefore the value-weighted return of decile 9 minus the value-weighted return of decile 0.

Exercise 3d: Performance metrics

The market benchmark is constructed as the value-weighted average excess return across all stocks in the relevant sample. This makes the benchmark comparable to the long-short portfolio, since both returns are measured in excess of the risk-free rate. CAPM alpha is estimated by regressing the long-short portfolio excess return on the value-weighted market excess return.

Table 4 reports annualized mean excess returns, Sharpe ratios, CAPM alphas, and alpha t-statistics for the long-short portfolios and the market benchmark. Elastic Net performs consistently across validation and test, with positive annualized returns and Sharpe ratios in both periods. Random

Forest performs reasonably in validation but delivers negative test-period performance, suggesting that its predictive signal is less stable out-of-sample. The alpha t-statistics are below conventional significance levels, so the evidence for statistically significant abnormal returns is weak.

Table 4: Annualized long-short portfolio performance and market benchmark

Model	Sample	ann_mean	sharpe	alpha	alpha_t
Elastic Net	Validation	0.041	0.6976	0.0405	1.3369
Random Forest	Validation	0.0364	0.3143	0.0702	1.2036
Market	Validation	0.1361	1.3293	nan	nan
Elastic Net	Test	0.0641	0.7228	0.0449	1.0329
Random Forest	Test	-0.0321	-0.1777	-0.0369	-0.4066
Market	Test	0.1748	1.0233	nan	nan

Exercise 3e: Portfolio turnover

Table 5 reports average monthly turnover.

Table 5: Average monthly portfolio turnover

Model	Validation	Test
Elastic Net	0.0073	0.0085
Random Forest	0.5418	0.5006

Table 5 shows that Elastic Net has lower turnover than Random Forest in the test period. Elastic Net turnover is 0.008, compared with 0.501 for Random Forest. This matters because high turnover increases transaction costs and can erode gross portfolio returns. Random Forest's turnover of around 50% per month implies substantial trading costs that would further reduce its already negative test-period return, whereas Elastic Net's turnover below 1% means its positive return is largely unaffected by trading frictions.

Exercise 3f: Model comparison

For the selected hyperparameter specifications, Elastic Net is preferred from an investor perspective based on test-period risk-adjusted performance. In the test period, Elastic Net generates a positive annualized mean return of 6.41%, a Sharpe ratio of 0.72, and a positive CAPM alpha of 4.49%. In comparison, Random Forest generates a negative annualized mean return of -3.21%, a Sharpe ratio of -0.18, and a negative CAPM alpha of -3.69%.

Overall, Elastic Net provides the better trade-off between risk-adjusted performance, alpha, and implementability in our sample.

Exercise 4

Exercise 4a: Variable importance

Variable importance is measured as the drop in annualized test Sharpe ratio when one predictor is set to a reference value and the long-short portfolio is rebuilt. For standardized continuous predictors, the reference value is zero, corresponding to the training-sample mean after standardization. For industry dummies, the reference value is set to the test-sample mean.

Table 6: Top predictors by Sharpe-ratio drop in the test sample

model	feature	sharpe_drop
Elastic Net	characteristic_mom1m	0
Elastic Net	characteristic_mom12m	0
Elastic Net	characteristic_mvel1	0
Random Forest	characteristic_ep_x_macro_bm	0.4633
Random Forest	characteristic_retvoll_x_macro_tbl	0.2944
Random Forest	characteristic_mvel1_x_macro_dfy	0.2885

This portfolio-based importance measure evaluates whether a predictor matters for the final investment strategy rather than only for return prediction accuracy. A larger Sharpe-ratio drop indicates that removing the information in that predictor reduces the economic value of the model.

Exercise 4b: Ranking comparison

The rankings differ across the two models because Elastic Net and Random Forest use the predictors in different ways. Elastic Net is a regularized linear model, so predictor importance is tied to linear effects after shrinkage. Random Forest is more flexible and can assign importance to predictors that matter through nonlinear patterns or interactions. Differences in the rankings therefore reflect differences in model structure, not only differences in predictive strength.

Exercise 4c: Findings to financial theory

The Random Forest importance results suggest that portfolio performance is mainly driven by state-dependent valuation, volatility, and size effects. The largest Sharpe-ratio drop comes from `characteristic_ep_x_macro_bm`, indicating that the predictive value of earnings-to-price depends on aggregate book-to-market conditions. This is consistent with value-based return predictability varying across market valuation states.

The other top predictors, `characteristic_retvoll_x_macro_tbl` and `characteristic_mvel1_x_macro_dfy`, suggest that volatility and size effects depend on interest rate conditions and credit risk. In contrast, Elastic Net produces near-zero importance scores, consistent with its strongly regularized linear structure. Overall, the results indicate that the Random Forest model extracts economic value primarily from nonlinear and macro-dependent interactions.

For Assignment 3

```
['ma2_random_forest_bundle.joblib']
```

References

- Goyal, A. and Welch, I. (2008). A comprehensive look at the empirical performance of equity premium prediction. *The Review of Financial Studies*, 21(4):1455–1508.
- Gu, S., Kelly, B., and Xiu, D. (2020). Empirical asset pricing via machine learning. *The Review of Financial Studies*, 33(5):2223–2273.